

Project Title

Privacy preserving collaborative resource and task orchestrator for research environments

Background

Research groups and academic labs often struggle with fragmented coordination. While tools like Trello or Slack exist, they often store sensitive research data on third-party servers, posing privacy risks for proprietary or pre-publication findings. Furthermore, existing tools rarely integrate specialized academic needs, such as tracking shared lab equipment usage, managing manuscript versioning across a team, or coordinating meeting minutes with automated action-item extraction. Small research teams often resort to a messy mix of spreadsheets and private messages, leading to "information silos" and missed deadlines.

Problem Description

Academic and small-scale professional teams face significant overhead in synchronizing tasks across diverse projects while maintaining strict data privacy. Key challenges include:

- Data vulnerability: sensitive research intellectual property is often hosted on public cloud platforms without end-to-end encryption.
- Resource conflicts: shared assets (like high-performance workstations, specialized sensors, or meeting rooms) are often double-booked due to a lack of a centralized scheduling system.
- Fragmented context: discussions about a specific task are often lost in chat history rather than being linked directly to the project's documentation or codebase.
- Complexity vs. utility: enterprise-grade project management suites are often too bloated and expensive for a 5–10 person research lab.

Proposed Solution

This project proposes a full-stack, self-hostable SaaS platform designed for secure team collaboration. The system will prioritize a "Privacy-First" architecture where all data is encrypted at rest and during transit. It will provide a unified interface for task boards, shared resource scheduling, and a centralized documentation hub.

A core innovation will be the Automated Context Engine, a separate backend service that processes team interactions to suggest task deadlines and summarize meeting notes. This ensures that even without manual entry, the system remains updated with the team's latest progress.

Objectives

- Analyze the coordination bottlenecks within small, high-output technical teams.
- Design a secure, multi-tenant architecture that supports end-to-end encrypted data storage.
- Implement a responsive web and mobile-friendly interface for real-time task tracking.
- Develop a resource-locking algorithm to manage shared hardware/software assets effectively.
- Evaluate the system's impact on reducing "admin time" for researchers through automated summaries.

Scope of the Project

The project involves complete full-stack implementation including a web frontend, a mobile companion app, and a microservices-based backend and visualization (graphs).

Included in Scope: (with example technologies, you can use your own technologies with a feasibility analysis)

- Web Application:
 - Interactive Kanban boards for task management.
 - Dashboard for resource scheduling (e.g., booking a lab GPU).
 - Rich-text editor for collaborative meeting minutes.
- Mobile Application:
 - Real-time push notifications for task updates and mentions.
 - Quick-capture interface for logging experimental notes on the go.
- Backend Services:
 - Core API: Handles user authentication, project permissions, and data persistence.
 - Context Service: An independent service using LangChain to summarize long discussion threads and extract action items.
 - Socket Server: Manages real-time collaborative editing and instant updates across clients.

Optional Scope:

- Integration with external 3rd-party cloud storage (e.g., Google Drive/Dropbox) to maintain privacy.
- Financial/Grant budget tracking modules.
- Heavyweight video conferencing tools (focus remains on text and task data).

Expected Outcomes

- A deployed prototype (say in Nexus Repository Community Edition platform).
- An automated summary engine that converts chat logs into structured task updates.
- Comprehensive technical documentation covering the encryption implementation and API schema.
- A performance report comparing the overhead of private encryption vs. standard cloud storage.